

**MOTOROLA**

2.0GHz Dual Modulus Prescaler

The MC12034A can be used with CMOS synthesizers requiring positive edges to trigger internal counters such as Motorola's MC145xxx series in a PLL to provide tuning signals up to 2.0 GHz in programmable frequency steps.

The MC12034B can be used with CMOS synthesizers requiring negative edges to trigger internal counters such as Fujitsu's MB87001.

A Divide Ratio Control (SW) permits selection of a 32/33 or 64/65 divide ratio as desired.

The Modulus Control (MC) selects the proper divide number after SW has been biased to select the desired divide ratio.

- 2.0 GHz Toggle Frequency
- Supply Voltage 4.5 to 5.5 V
- MC12034A for Positive Edge Triggered Synthesizers
- 12mA Maximum, -40 to 85°C , $V_{CC} = 5.5$ Vdc
- Modulus Control Input is Compatible with Standard CMOS and TTL
- Low-Power 8.5 mA Typical

FUNCTIONAL TABLE

SW	MC	Divide Ratio
H	H	32
H	L	33
L	H	64
L	L	65

NOTES: 1. SW: H = V_{CC} , L = Open. A logic L can also be applied by grounding this pin, but this is not recommended due to increased power consumption.
2. MC: H = 2.0 V to V_{CC} , L = GND to 0.8 V.

Design Criteria	Value	Unit
Internal Gate Count *	67	ea
Internal Gate Propagation Delay	200	ps
Internal Gate Power Dissipation	0.75	mW
Speed Power Product	0.15	pJ

NOTE: *Equivalent to a two-input NAND gate.

MAXIMUM RATINGS

Characteristic	Symbol	Range	Unit
Power Supply Voltage, Pin 2	V_{CC}	-0.5 to $+7.0$	Vdc
Operating Temperature Range	T_A	-40 to $+85$	$^{\circ}\text{C}$
Storage Temperature Range	T_{stg}	-65 to $+150$	$^{\circ}\text{C}$
Modulus Control Input, Pin 6	MC	-0.5 to $+6.5$	Vdc

NOTES: 1. ESD data available upon request.

2. This device contains protection circuitry to guard against damage due to high static voltages or electric fields. However, precautions must be taken to avoid applications of any voltage higher than maximum rated voltages to this high impedance circuit. For proper operation, V_{in} and V_{out} should be constrained to the range $\text{GND} \leq (V_{in} \text{ or } V_{out}) \leq V_{CC}$.

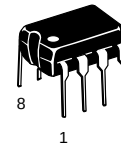
MC12034A MC12034B

MECL PLL COMPONENTS $\div 32/33$, $\div 64/65$ DUAL MODULUS PRESCALER

SEMICONDUCTOR TECHNICAL DATA

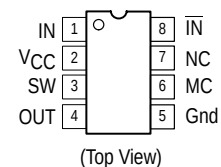


D SUFFIX
PLASTIC PACKAGE
CASE 751
(SO-8)



P SUFFIX
PLASTIC PACKAGE
CASE 626

PIN CONNECTIONS



ORDERING INFORMATION

Device	Operating Temp Range	Package
MC12034AD	$T_A = -40^{\circ}$ to $+85^{\circ}\text{C}$	SO-8
MC12034AP		Plastic
MC12034BD		SO-8
MC120348BP		Plastic

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ELECTRICAL CHARACTERISTICS ($V_{CC} = 4.5$ to 5.5 Vdc, $T_A = -40$ to 85°C , unless otherwise noted.)

Characteristic	Symbol	Min	Typ	Max	Unit
Toggle Frequency (Sine Wave)	f_t	0.5	2.4	2.0	GHz
Supply Current Output Unloaded (Pin 2)	I_{CC}	—	8.5	12	mA
Modulus Control Input High (MC)	V_{IH1}	2.0	—	V_{CC}	V
Modulus Control Input Low (MC)	V_{IL1}	—	—	0.8	V
Divide Ratio Control Input High (SW)	V_{IH2}	V_{CC}	V_{CC}	V_{CC}	Vdc
Divide Ratio Control Input Low (SW)	V_{IL2}	OPEN	OPEN	OPEN	—
Output Voltage Swing ($C_L = 12$ pF, $R_L = 1.1$ k Ω)	V_{out}	1.0	1.6	—	V _{pp}
Modulus Setup Time MC to Out	t_{SET}	—	8.0	10.0	ns
Input Voltage Sensitivity 500–2000 MHz	V_{in}	100	—	1500	mV _{pp}
Output Current ($C_L = 12$ pF, $R_L = 1.1$ k Ω)	I_O	—	—	3.5	mA

Figure 1. Logic Diagram (MC12034A)

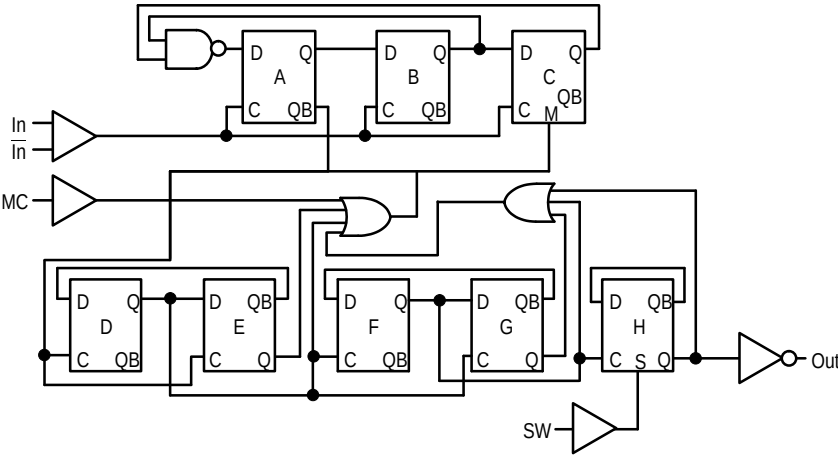


Figure 2. Modulus Setup Time

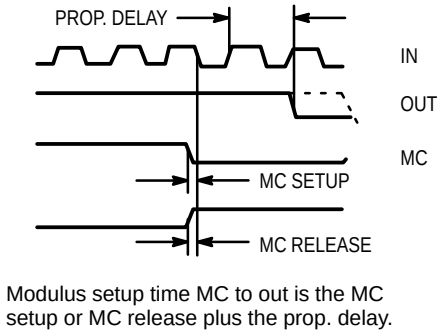
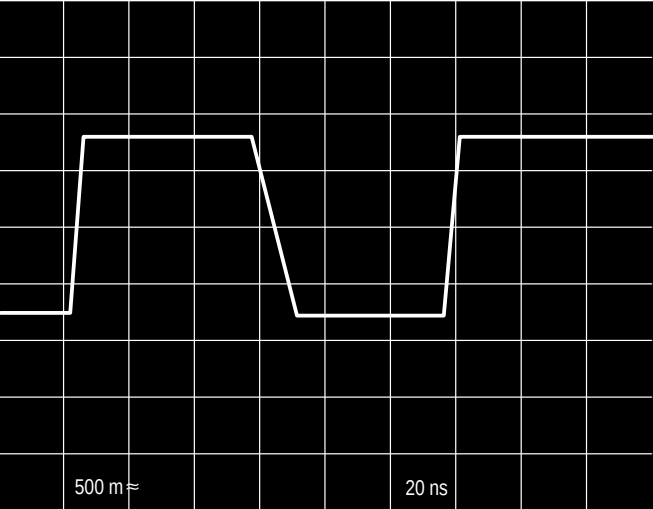


Figure 3. Typical Output Waveform



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Figure 4. AC Test Circuit

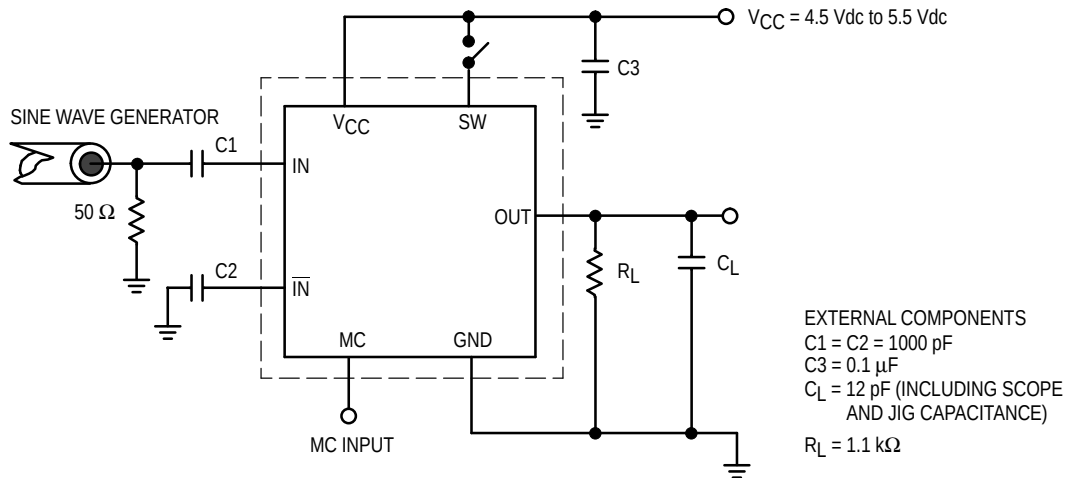


Figure 5. Input Signal Amplitude versus Input Frequency

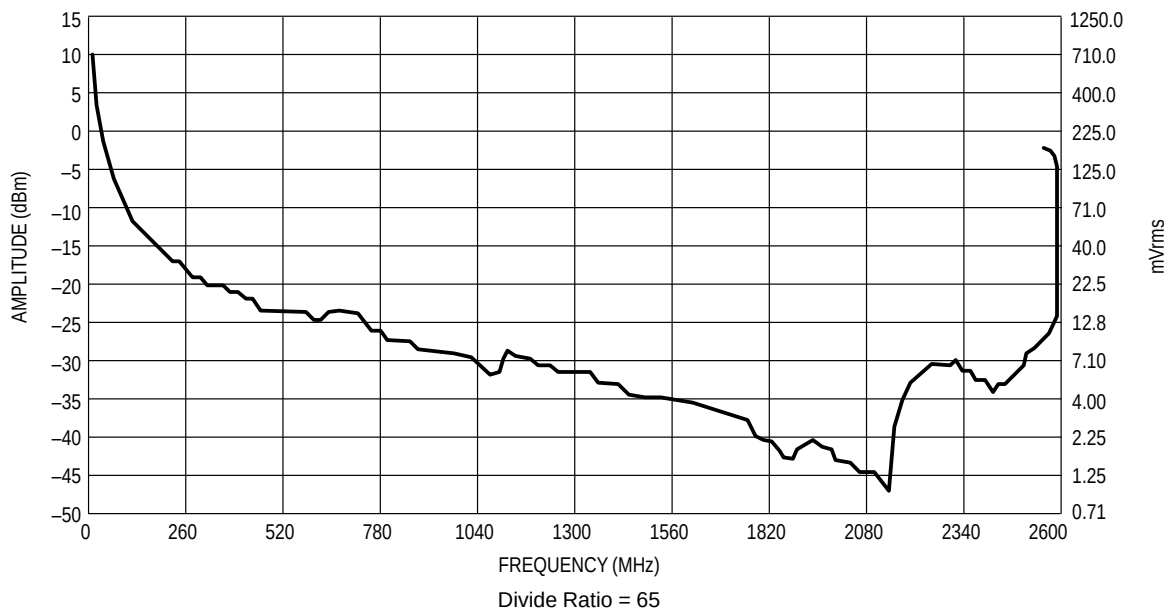
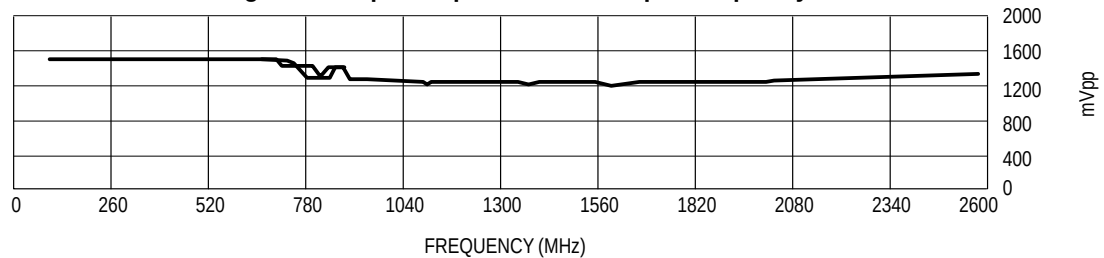
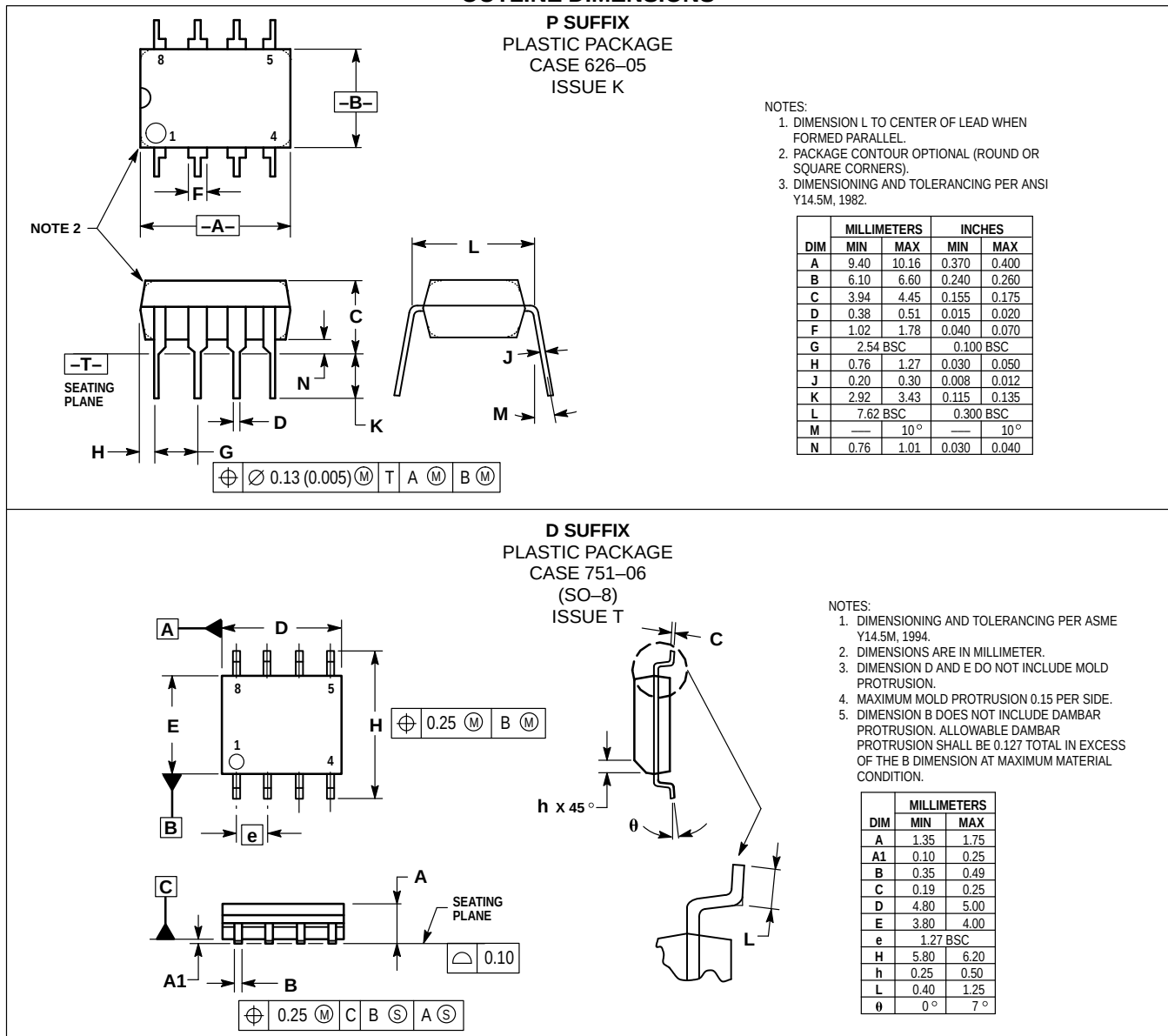


Figure 6. Output Amplitude versus Input Frequency



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OUTLINE DIMENSIONS



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